Web Development Foundations

Week 7 – "From Working – To Clean"

Day 2: "Function Refactor Iteration"

1

Reconnect

Monday:

- working code can still be hard to read
- long event handlers hide responsibilities
- functions name responsibilities
- refactoring must preserve behavior

Today:

- We perform the refactor one step at a time.

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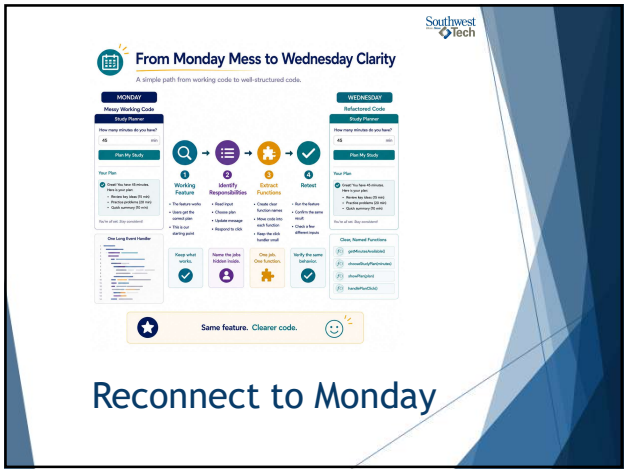

Reconnect to Monday

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The Same Feature, Clearer Code

- **The feature:** *enter minutes and receive a study plan.*
- **The improvement:** *separate the jobs.*
- **Before:** *one event handler does everything.*
- **After:** *named functions each do one clear job.*

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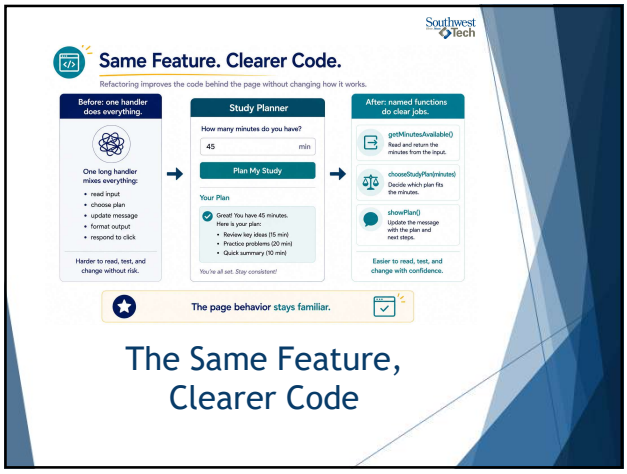

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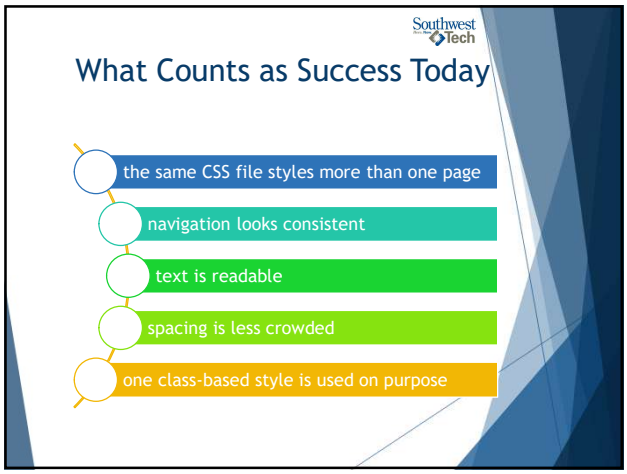

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Refactor In Small Steps

Do not refactor everything at once.

Use this rhythm:

1. choose one responsibility
2. move that code into a function
3. reconnect the call
4. retest
5. then choose the next responsibility

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Small-Step Refactor

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What We Will Use Today

- expected behavior
- named function
- return value
- parameter
- event listener
- callback
- arrow function awareness
- retest

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Today's Refactoring Toolbox
Tools for turning messy working code into clear, well-structured code.
expected behavior
named function
return value
parameter
event listener
callback
arrow function awareness
retest
Use the right tool. One step at a time.
Refactoring Toolbox

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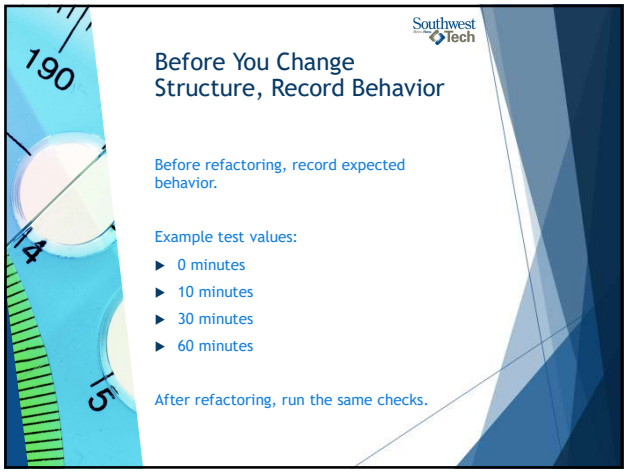
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Before You Change Structure, Record Behavior
Before refactoring, record expected behavior.
Example test values:
0 minutes
10 minutes
30 minutes
60 minutes
After refactoring, run the same checks.

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Record Behavior Before Refactoring
Capture what the feature does so you can confirm it stays the same.
Before Refactor
Test These Values
0 minutes
10 minutes
30 minutes
60 minutes
Behavior recorded and confirmed. This is our baseline.
Use the same values and expected behavior after refactoring.
After Refactor
Retest the Same Values
0 minutes
10 minutes
30 minutes
60 minutes
Behavior matches the baseline. The feature still works the same.
Retest the same values.
Record Behavior First

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Extract One Responsibility

Look for one job:

- read input
- validate input
- choose a plan
- update the page
- handle the click

Each function should have a name that says its job.

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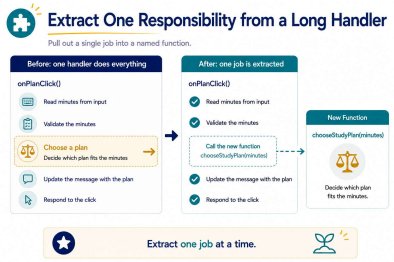
Extract Responsibility



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Return Values Carry Results Back

A return value sends a result back.

Example:
`` `chooseStudyPlan(minutes)` ``
returns the message to show.

Then another function can use that result.
Returning a value keeps jobs separate.

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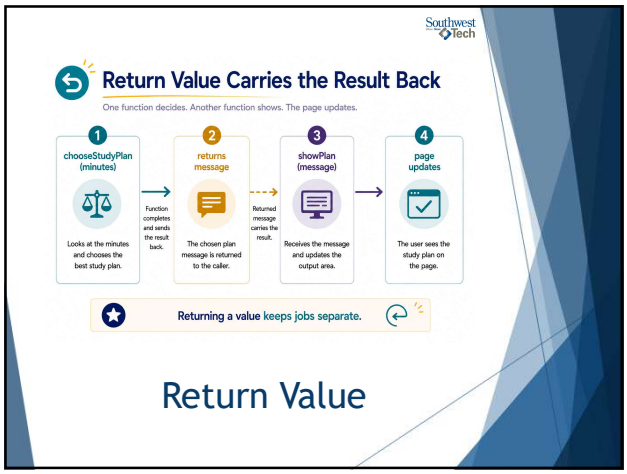
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Return Value Carries the Result Back
One function decides. Another function shows. The page updates.
1
chooseStudyPlan (minutes)
Looks at the minutes and chooses the best study plan.
Function completes and sends the result back.
2
returns message
The chosen plan message is returned to the caller.
Returned message carries the result.
3
showPlan (message)
Receives the message and updates the output area.
4
page updates
The user sees the study plan on the page.
Returning a value keeps jobs separate.
Return Value

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Demo
"Refactor Into Named Functions"

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What to Watch For
record expected behavior
extract `getMinutesAvailable()`
extract `chooseStudyPlan(minutes)`
extract `showPlan()`
reconnect the click listener
retest the same values
Q2
Q3

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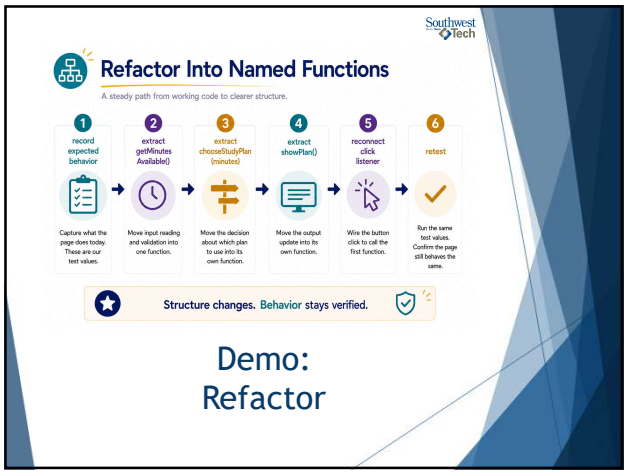
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Refactor Into Named Functions
A steady path from working code to clearer structure.
1 record expected behavior
Capture what the page does today. These are our test values.
2 extract getMinutesAvailable()
Move input reading and validation into one function.
3 extract chooseStudyPlan (minutes)
Move the decision about which plan to use into its own function.
4 extract showPlan()
Move the output update into its own function.
5 reconnect click listener
Wire the button click to call the first function.
6 retest
Run the same test values. Confirm the page still behaves the same.
Structure changes. Behavior stays verified.
Demo: Refactor

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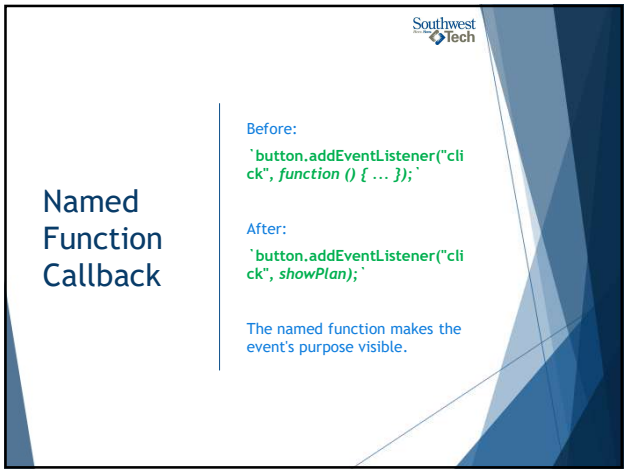
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Named Function Callback
Before:
`button.addEventListener("click", function () { ... });`
After:
`button.addEventListener("click", showPlan);`
The named function makes the event's purpose visible.

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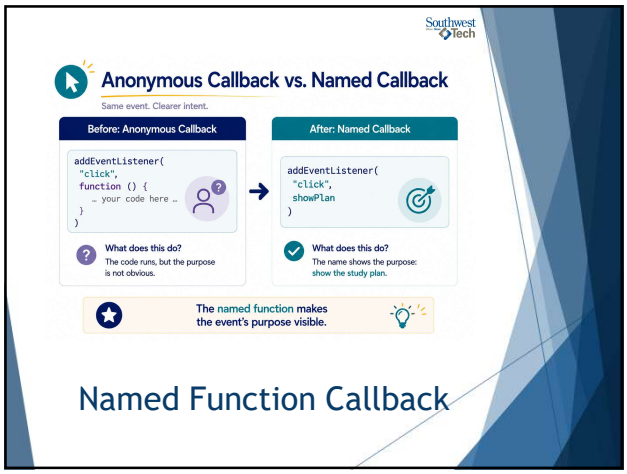
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Anonymous Callback vs. Named Callback
Same event. Clearer intent.
Before: Anonymous Callback
addEventListener("click", function () { … your code here … })
What does this do?
The code runs, but the purpose is not obvious.
After: Named Callback
addEventListener("click", showPlan)
What does this do?
The name shows the purpose: show the study plan.
The named function makes the event's purpose visible.
Named Function Callback

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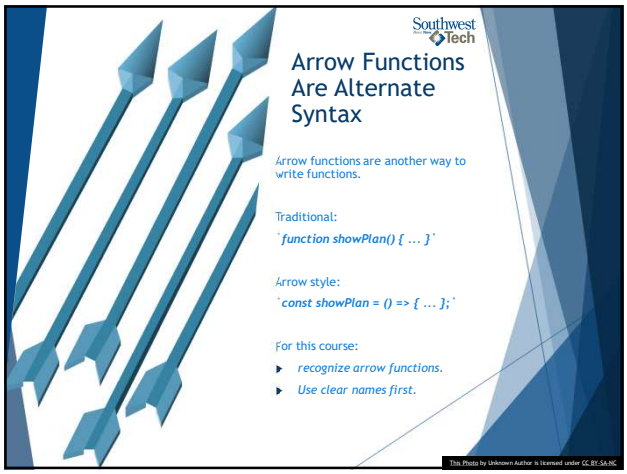
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Arrow Functions Are Alternate Syntax
Arrow functions are another way to write functions.
Traditional:
`function showPlan() { ... }`
Arrow style:
`const showPlan = () => { ... };`
For this course:
recognize arrow functions.
Use clear names first.
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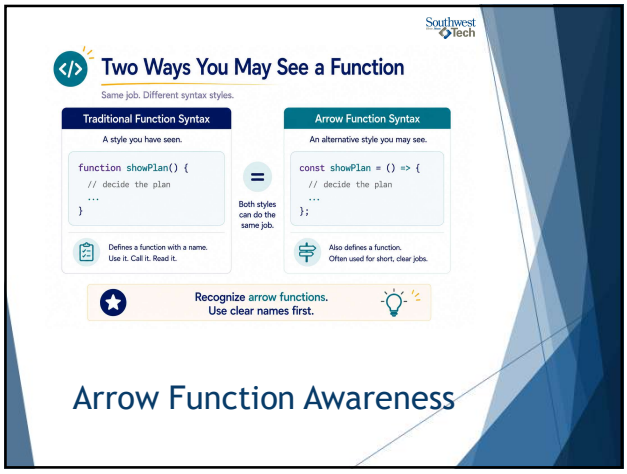
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Two Ways You May See a Function
Same job. Different syntax styles.
Traditional Function Syntax
A style you have seen.
function showPlan() {
// decide the plan
...
}
Defines a function with a name. Use it. Call it. Read it.
Both styles can do the same job.
Arrow Function Syntax
An alternative style you may see.
const showPlan = () => {
// decide the plan
...
};
Also defines a function. Often used for short, clear jobs.
Recognize arrow functions. Use clear names first.
Arrow Function Awareness

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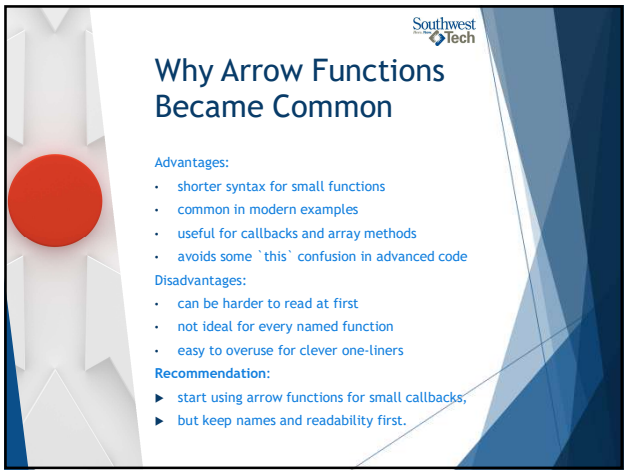
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Why Arrow Functions Became Common
Advantages:
shorter syntax for small functions
common in modern examples
useful for callbacks and array methods
avoids some `this` confusion in advanced code
Disadvantages:
can be harder to read at first
not ideal for every named function
easy to overuse for clever one-liners
Recommendation:
start using arrow functions for small callbacks,
but keep names and readability first.

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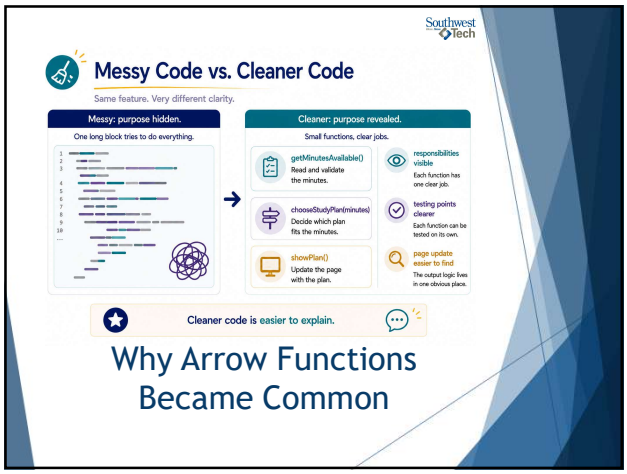
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Messy Code vs. Cleaner Code
Same feature. Very different clarity.
Messy: purpose hidden.
One long block tries to do everything.
Cleaner: purpose revealed.
Small functions, clear jobs.
getMinutesAvailable()
Read and validate the minutes.
chooseStudyPlan(minutes)
Decide which plan fits the minutes.
showPlan()
Update the page with the plan.
responsibilities visible
Each function has one clear job.
testing points clearer
Each function can be tested on its own.
page update easier to find
The output logic lives in one obvious place.
Cleaner code is easier to explain.
Why Arrow Functions Became Common

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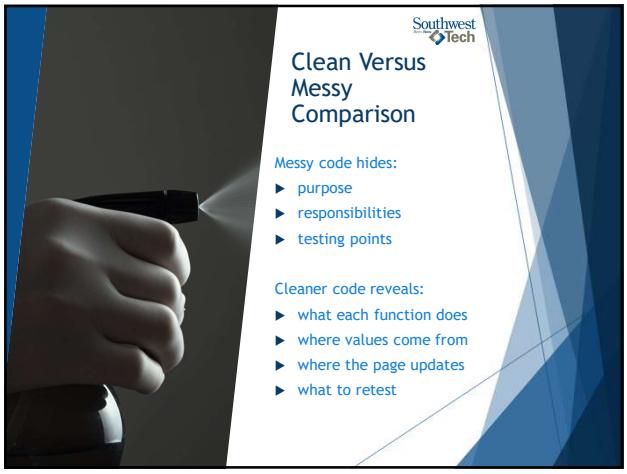
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Clean Versus Messy Comparison
Messy code hides:
purpose
responsibilities
testing points
Cleaner code reveals:
what each function does
where values come from
where the page updates
what to retest

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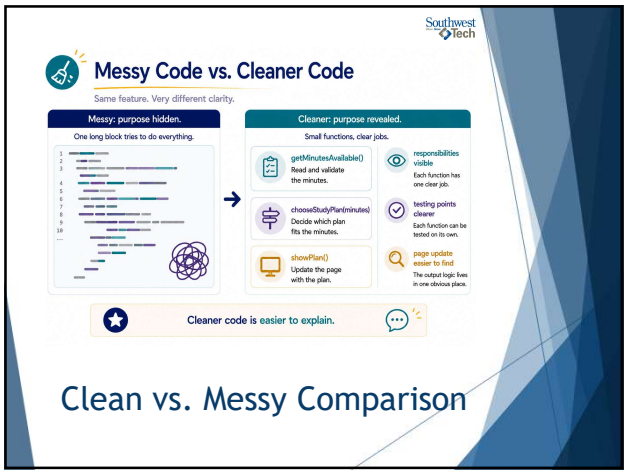
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Clean vs. Messy Comparison

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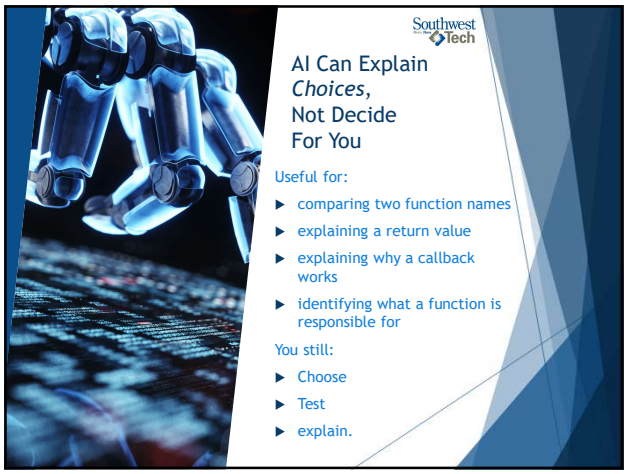
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AI Can Explain *Choices*, Not Decide For You
Useful for:
comparing two function names
explaining a return value
explaining why a callback works
identifying what a function is responsible for
You still:
Choose
Test
explain.

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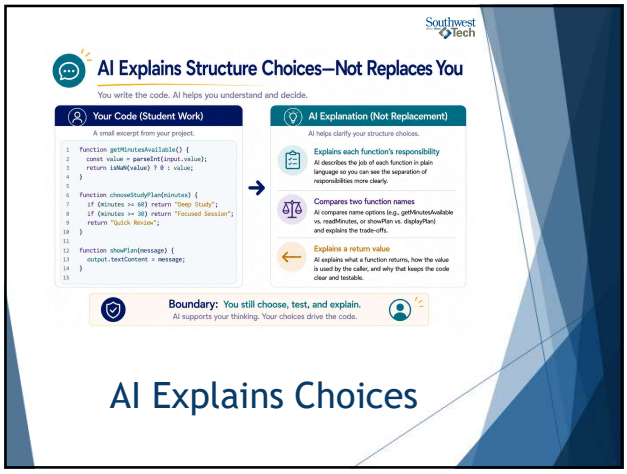
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AI Explains Structure Choices—Not Replaces You
You write the code. AI helps you understand and decide.
Your Code (Student Work)
A small excerpt from your project.
AI Explanation (Not Replacement)
AI helps clarify your structure choices.
Explains each function's responsibility
AI describes the job of each function in plain language so you can see the separation of responsibilities more clearly.
Compares two function names
AI compares name options (e.g., getMinutesAvailable vs. readMinutes, or showPlan vs. displayPlan) and explains the trade-offs.
Explains a return value
AI explains what a function returns, how the value is used by the caller, and why that keeps the code clear and testable.
Boundary: You still choose, test, and explain.
AI supports your thinking. Your choices drive the code.
AI Explains Choices

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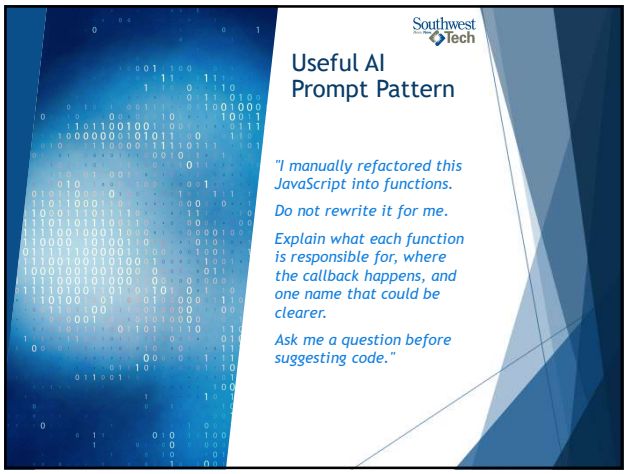
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Useful AI Prompt Pattern
"I manually refactored this JavaScript into functions.
Do not rewrite it for me.
Explain what each function is responsible for, where the callback happens, and one name that could be clearer.
Ask me a question before suggesting code."

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AI Prompt Pattern

For Explaining a JavaScript Refactor

Context	I manually refactored this JavaScript into functions.
Constraint	Do not rewrite it for me.
Explain	Please explain the code's: • responsibilities • callback • naming choices
Ask First	Ask me a question before suggesting code.

Prompt for explanation, not replacement.

Useful AI Prompt Pattern

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Lab Refinement

Refine structure

Your goal:

- improve function names
- organize logic clearly
- reduce repetition where possible
- keep the feature working
- add one clarity improvement

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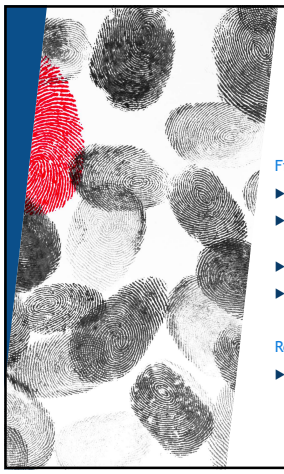

Evidence & Reflection

Final Assignment 7 evidence:

- updated HTML, CSS, and JS
- JavaScript uses functions effectively
- function names are meaningful
- behavior still works

Reflection

- Explain what changed

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How To Read Next Week's Material

Required:

- read the async handout for the big idea
- focus on what happens now vs what happens later

Skim:

- Ajax and JSON as background
- JSON structure guide for objects and arrays

Reference:

- MDN links are there when you need exact syntax

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Structure To Async Bridge

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Closing Success Check

This week:

- working code became clearer
- responsibilities became functions
- callbacks became easier to read
- behavior stayed verified

Next:

- systems respond over time.

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